

C01

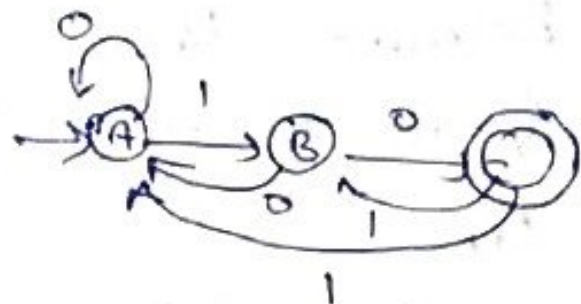
1. Given NFA

States:

$$Q = \{A, B, C\}$$

Alphabet

$$\Sigma = \{0, 1\}$$



Initial state: A, Final state: C

from the diagram

$$\delta(A, 0) = A$$

$$\delta(A, 1) = B$$

$$\delta(B, 0) = A$$

$$\delta(B, 1) = \emptyset$$

$$\delta(C, 0) = B$$

$$\delta(C, 1) = A$$

There is no ϵ -transitions

Method used: Subset construction

Basic idea of subset construction is

$\Rightarrow$ Each state of the DFA represents a set of possible states of NFA.

The possible DFA states are:

$$\{A\}, \{B\}, \{C\}, \emptyset$$

Here, $\emptyset$ is the dead state.
Because there is no transition
from B on input, 1.

Let

$$q_0 = \{A\}$$

$$q_1 = \{B\}$$

$$q_2 = \{C\}$$

$$q_d = \emptyset$$

initial DFA state = $\{A\}$

final DFA state = $\{C\}$

DFA transition calculation

from $\{A\}$ on 0

$$\delta(\{A\}, 0) = \{A\}$$

on 1:

$$\delta(\{A\}, 1) = \{B\}$$

from $\{B\}$ on 0:

$$\delta(\{B\}, 0) = \{A\}$$

on 1:

$$\delta(\{B\}, 0) = \{A\} \quad \text{on 1:}$$

$$\delta(\{B\}, 1) = \emptyset$$

From $\{c\}$: on 0

$$\delta(\{c\}, 0) = \{B\}$$

on 1:

$$\delta(\{c\}, 1) = \{A\}$$

From $\emptyset$ on 0

$$\delta(\emptyset, 0) = \emptyset$$

on 1:

$$\delta(\emptyset, 1) = \emptyset$$

DFA	Input 0	Input 1
$\{A\}$	$\{A\}$	$\{B\}$
$\{B\}$	$\{A\}$	$\emptyset$
$\{c\}$	$\{B\}$	$\{A\}$
$\emptyset$	$\emptyset$	$\emptyset$

Challenges in NFA to DFA conversion

1. An NFA can have multiple possible states for the same input, whereas a DFA must have exactly one state for each input.
2. Therefore, a set of NFA states is represented as one DFA state

3. All possible combination of NFA have to be considered

2.

Given Statement

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Let $P(n)$ be statement

$$P(n): 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Base Case:

$$n = 1$$

L.H.S

$$\sum_{i=1}^1 i^2 = 1^2 = 1$$

R.H.S

$$\frac{n(n+1)(2n+1)}{6}$$

$$\text{Sub } n=1$$

$$= \frac{1(1+1)(2(1)+1)}{6}$$

$$= \frac{1 \times 2 \times 3}{6}$$

$$= 1$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

$P(1)$ is true

$\therefore$ The statement is true for $n = 1$

Now assume $n = k$

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

To prove it for $P(k+1)$ is true

L.H.S

(b) ~~3/12/2~~ $1^2 + 2^2 + \dots + k^2 + (k+1)^2$
using induction

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$(k+1) \left[\frac{k(2k+1)}{6} + (k+1) \right]$$

$$= \frac{(k+1)(2k^2 + k + 6k + 6)}{6}$$

$$= \frac{(k+1)(2k^2 + 7k + 6)}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

R.H.S

$$n = k+1$$

$$= \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{(k+1)[(k+1)+1][2(k+1)+1]}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

Since

$$L.H.S = R.H.S$$

True

∴ By the principle of mathematical induction

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Hence proved.

